

- 3–10 (a)** Formulate mesh-current equations for the circuit in Figure P3–10.
- (b)** Use these equations to find v_x and i_x .

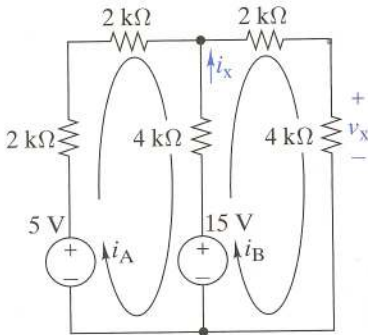


FIGURE P3–10

3-13 (a) Formulate mesh-current equations for the circuit in Figure P3-13.

(b) Solve for v_x and i_x when $R_1 = R_2 = 10\text{ k}\Omega$, $R_3 = 2\text{ k}\Omega$, $R_4 = 1\text{ k}\Omega$, $i_s = 2.5\text{ mA}$, $v_{S1} = 12\text{ V}$, and $v_{S2} = 0.5\text{ V}$.

(c) Find the power supplied by v_{S1} .

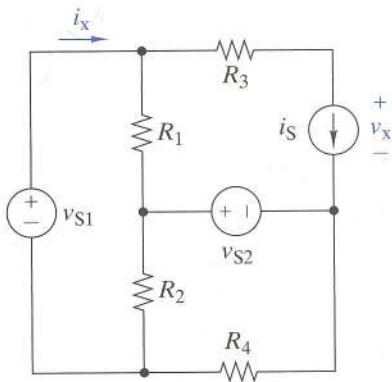


FIGURE P3-13

- 3-15 (a) Formulate mesh-current equations for the circuit in Figure P3-15.
- (b) Use these equations to find v_x and i_x .
- (c) Find the total power delivered to the resistors.

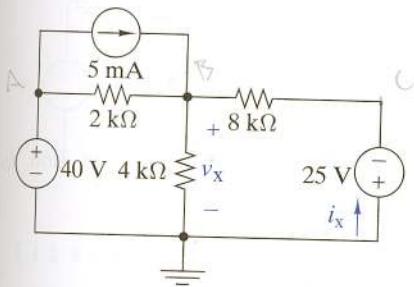


FIGURE P3-15

P3-17 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

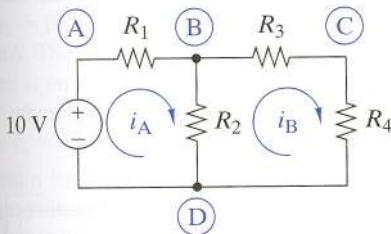


FIGURE P3-17

3–26 Use the superposition principle to find v_O in Figure P3–26.

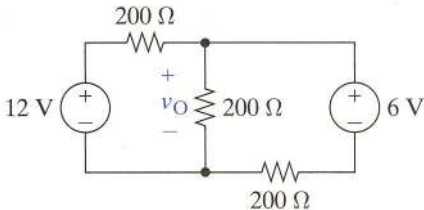


FIGURE P3–26

3-31 Use the superposition principle to find v_O in terms of i_1 , i_2 , and R in Figure P3-31.

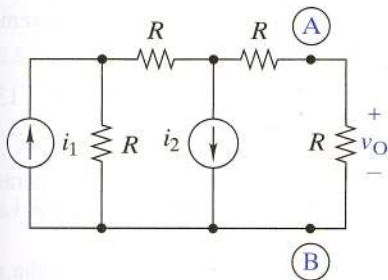


FIGURE P3-31

3-32 A linear circuit containing two sources drives a $100\text{-}\Omega$ load resistor. Source number 1 delivers 250 mW to the load when source number 2 is off. Source number 2 delivers 4 W to the load when source number 1 is off. Find the power delivered to the load when both sources are on. *Hint:* The answer is not 4.25 W . Why?

4-1 Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4-1 for $r = 4 \text{ k}\Omega$.

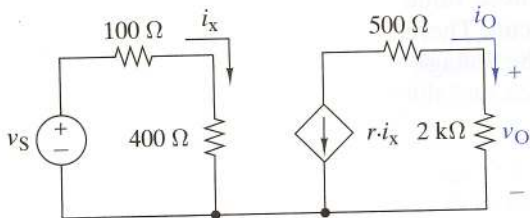


FIGURE P4-1

4-5 Find the current gain i_O/i_S in Figure P4-5.

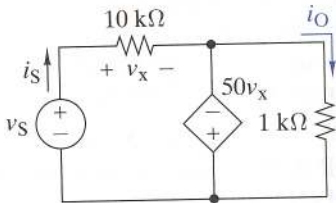


FIGURE P4-5

4-6 Find an expression for the current gain i_O/i_S in Figure P4-6. Hint: Apply KCL at node A.

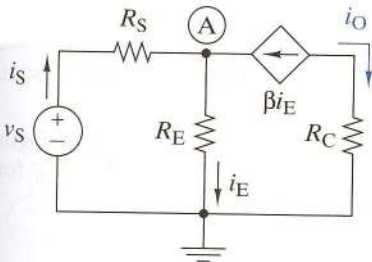


FIGURE P4-6